

Bank Customer Data Analysis

Project Overview

This project is a simple **Bank Customer Data Analysis** project created using Python. The main purpose of this project is to generate a synthetic bank customer dataset and perform basic data analysis and visualization on it.

The dataset contains **200 customer records** with information such as Customer ID, Name, Age, Gender, Income, and Account Balance.

Objective

- Generate fake bank customer data.
- Check and analyze the data.
- Find missing values and basic statistics.
- Create graphs to understand customer data.
- Find average income by gender.
- Find the top 5 customers by account balance.

Technologies & Libraries

- Python – Main programming language
- Pandas – Data handling and analysis
- NumPy – Random numerical data generation
- Faker – Creating fake customer information
- Matplotlib – Data visualization
- CSV – Storing the dataset

Project Workflow

Generate Data → Save CSV → Load Data → Check Data → Analyze Data → Visualize Data

Conclusion

This project helped me understand how Python libraries can be used together for a basic data analysis project. I learned how to create synthetic data, store it in a CSV file, load and analyze it using Pandas, generate numerical data using NumPy, create fake data using Faker, and visualize data using Matplotlib. The project provides a basic understanding of the data analysis workflow, from **data generation and data checking to analysis and visualization**.